MODULE G

Machine learning and AI engineering: a learning roadmap

A staged path from Python to deployment. Each stage ends with a small project.

Stage	Learn	Project
1. Python	Variables, lists, functions, files	Clean a CSV file
2. Maths and statistics	Averages, spread, probability, correlation	Describe a sales dataset
3. NumPy and pandas	Arrays, tables, grouping, missing values	Build a customer summary
4. Scikit-learn	Train, test, metrics, feature engineering	Lead-scoring model (see the ML engineer chapter)
5. Deep learning (PyTorch or TensorFlow)	Neural networks, training loops	Image classifier on a public dataset
6. NLP and computer vision	Text classification, image tasks	Ticket classifier
7. Deployment and MLOps	APIs, monitoring, retraining	Model behind an API with logging

Official resources to check: scikit-learn, PyTorch tutorials, TensorFlow tutorials, Kaggle Learn. Links not checked live.

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